GANs for channel data generation

🎯 Objective: Create a GAN that generates channel impulse responses based on our measurements
🗓️ Temptative due date: June 2026

Motivation

In wireless communications, the design and evaluation of transmission systems heavily rely on understanding and modeling propagation channels. These channels represent how signals travel from the transmitter to the receiver and are influenced by distance, obstacles, reflection, diffraction, and noise. Accurate channel models ensure that communication systems work efficiently in real-world environments. However, obtaining real propagation channel measurements is costly and labor-intensive. It requires specialized equipment, field measurement time, and access to specific environments (urban, rural, indoor, etc.). Additionally, measurements are often limited to certain conditions, making generalization difficult. Theoretical channel models, while helpful, often oversimplify reality and fail to capture the complexity of dynamic and heterogeneous environments.

These ideas lead to the main motivation of this project: How can we generate synthetic propagation channels that are statistically similar to real ones without costly and extensive measurements? The answer could revolutionize wireless communication system design and testing, enabling engineers to simulate various propagation scenarios quickly and efficiently. The key technological challenge lies in the complexity of propagation channels and the difficulty of generating synthetic data that accurately captures their characteristics. However, Artificial Intelligence (AI) techniques, particularly Generative Adversarial Networks (GANs), can be transformative. GANs consist of a generator network that creates data and a discriminator network that evaluates it. The optimization process aims to make the synthetic data indistinguishable from real data by the discriminator.

Objectives

The main objective will be to create a synthetic wireless channel data generator model based on real measurements using GANs. To achieve this, the following partial objectives must be met:

- Analyze the current literature on GAN-based data generators
- Adapt existing models to our use case or develop new models

- Test their performance against other existing methods

Work plan

This work will follow the next work plan:

- WP1: Initial literature review (June-August)
- WP2: First implementation (September - November)
- WP3: Familiarization with wireless communications (December - February)
- WP4: Implementation under the wireless communications umbrella (January - March)
- WP5: Validation & Optimization (April - May)
- WP6: Documentation (June)

During the development of the project, the work plan may suffer modifications. The student and the supervisors will always decide on these modifications.

Resources

Team members

Team Member	Position	Expertise area
Dr./Prof. Eneko Iradier	Professor in the Computer Languages and Systems Dept.	AI/ML for wireless communications
Dr./Prof. Iñigo Bilbao	Professor in the Applied Mathematics Dept.	AI/ML for wireless communications
Dr./Prof. Pablo Angueira	Professor in the Communications Engineering Dept.	Radiocommunications and PHY layer
Dr./Prof. Jon Montalban	Professor in the Electronic Technology Dept.	Networking
Dr./Prof. Marta Fernandez	Professor in the Electronic Technology Dept.	Field trials

Others

Our laboratory is located in building I (P4B6). Several bachelor's, master's, and PhD students work daily, so students can join the laboratory on specific days or continuously.

We have desktop computers for simulations. Remote connections are available. We also have a server hosted in Leioa, where large simulation processes are carried out. If available, the students can use these computers.

We have other HW equipment oriented to field trials, such as USRPs, software-defined ratios, different standard communication kits, channel emulators, antennas, etc.

Deliverables

- Implementations in Python
- Intermediate reports/presentations
- Paper-style technical description and solution of the problem (4-8 pages)